UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF YEAR TWO SEMESTER TWO EXAMINATIONS, 2024/25

B.ED. PROGRAMME

COURSE CODE: **TEEG 212**

COURSE TITLE: **TEACHING AND ASSESSING NUMERACY 1 AT THE EARLY GRADE**

Instruction: Answer **all** questions in Section A by circling the correct option and any **three** in Section B on the answer booklet.

Time: 2 hours

SECTION A

[25 Marks]

Answer all the questions in this section.

1. After delivering a short maths lesson to classmates, a pre-service teacher listens to feedback about pacing and clarity from peers. She then reteaches the lesson, incorporating the suggestions. This process demonstrates _____.
   A. peer teaching
   B. assessment of learning
   C. assessment as learning
   D. peer testing of teaching

2. The main purpose of classroom assessment is to
   A. compare learner's performance.
   B. Improve instruction.
   C. measure teachers' effectiveness.
   D. report to parents.

3. Which of one the following curricula is currently being used in Ghanaian schools, that is shifting the structure and content of the education system from merely passing examinations to building character?
   A. National Council curriculum.
   B. Subject based curriculum.
   C. Objective-based curriculum.
   D. Standard-based curriculum.

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4. Content standard as used in the new mathematics curriculum refers to
   A. broad areas of the mathematics content to be studied.
   B. outcome or milestone that learners must exhibit for each lesson.
   C. topics within broad areas under which the content is organized.
   D. the pre-determined level of knowledge, skill, and attitudes that learners attain by a set stage of education.

5. The connected body of core skills that students are supposed to acquire throughout the process of teaching and learning in the early grade mathematics syllabus are referred to as
   A. Competencies.
   B. Indicators.
   C. Outcomes.
   D. Standards.

6. One reason why a teacher would engage children in sorting activities is to enable them _____
   A. add numbers.
   B. conserve numbers.
   C. identify numbers.
   D. subtract numbers.

7. A teacher asks learners to count the same set of beads in ones, twos, fives, and tens. This activity is most directly aimed at developing_____.
   A. Memorisation of counting patterns in the number systems
   B. Understanding of unitising and grouping in number systems
   C. Recognition of prime numbers that are used for counting in the number system
   D. Mastery of long division in the real number system of counting.

8. When a child is able to tell that the number of objects in a group does not change even if they are spread over a large space, the child is said to have conserved____
   A. Volume.
   B. Number.
   C. Iconic.
   D. Symbolic.

9. Which one of the following does a child need to know to be able to count the objects in a given group?
   A. Conservation of numbers.
   B. Sorting of objects.
   C. The numerals.
   D. The number names.

10. Which one of these is the most suitable order of activities designed to help children understand early number concepts?

A. Abstract, Concrete, and Semi-concrete.
B. Concrete, Abstract, and Semi-concrete.
C. Concrete, Semi-concrete and Abstract.
D. Semi-concrete, Abstract, Concrete.

11. A child who cannot count can determine which of two groups has more members by_____
    A. Counting.
    B. Matching.
    C. Sorting.
    D. Grouping.

12. A school's water tank holds 1,250 litres of water. Which of these is the correct place value breakdown?
    A. 1 thousand, 25 tens, 0 ones
    B. 12 hundreds, 50 ones
    C. 1 hundred, 2 tens, 5 one
    D. 1 thousand, 2 hundreds, 5 tens, 0 ones

13. Which of the following numerals has the 5 in the hundreds position?
    A. 9580132.
    B. 42494598.
    C. 104945230.
    D. 6841509210.

14. A bus travelled 2,430 km in total over a week. The expanded form of 2,430 is_____.
    A. $2 \times 1{,}000 + 43 \times 10 + 0 \times 1$
    B. $2 \times 1{,}000 + 400 \times 10 + 3 \times 1$
    C. $2 \times 1{,}000 + 4 \times 100 + 3 \times 10 + 0 \times 1$
    D. $2 \times 100 + 4 \times 1{,}000 + 3 \times 10 + 0 \times 1$

15. A procedure in teaching early grade mathematics where items or elements in a group are paired or associated with those of another group is termed
    A. Classification.
    B. Comparison.
    C. Matching.
    D. Sorting.

16. Which of Dienes' base ten blocks would you use to represent the digit "8" in helping a primary 2 learner to understand the value of the number 82?
    A. Blocks.
    B. Flats.
    C. Longs.
    D. Units.

17. When pre-service teachers practise delivering a short lesson to their classmates and then receive feedback, they are engaging in:
    A. Peer observation in real time.

B. Micro teaching for skill development
C. Summative assessment in teaching and learning.
D. Lecture-based instruction for teaching and learning.

18. A teacher models 7+57 + 5 by starting at 7 on a number line and counting forward 5 steps. This is an example of which strategy?
    A. Friendly jumps
    B. Counting on
    C. Making doubles
    D. Constant difference.

19. A bus has 48 passengers. At the next stop, 7 more passengers board. The driver thinks: "I'll add 2 to make 50, then add 5 more to make 55." This thinking illustrates:
    A. Constant difference used in counting.
    B. Friendly jumps to a decade number.
    C. Making doubles to make counting easier.
    D. Counting all objects at a time to save numbers.

20. Which one of the following is an objective of summative assessment?
    A. It attempts to correct errors and difficulties of pupils.
    B. It identifies and examines critically content to be assessed.
    C. It provides information about learner's level of achievement.
    D. It provides opportunity to assess pupils' writing skills in examination.

21. A farmer has 48 goats, and his neighbour has 35 goats. How many more goats does the farmer have?
    This method uses the:
    A. 83
    B. 48
    C. 35
    D. 13

22. A collection of pearls in three years is 172, 235 and 112 more than the collection of the first two years respectably. What is the total collection?
    A. 407
    B. 519
    C. 926
    D. 1038

23. A farmer has 36 goats and buys 25 more. A learner calculates:
    (30+20)+(6+5)=50+11=61(30 + 20) + (6 + 5) = 50 + 11 = 61. Which addition strategy does this show?
    A. Decomposing numbers into tens and ones
    B. Counting on from the smaller number
    C. Friendly jumps
    D. Constant difference

24. Which one of these skills demonstrates an understanding of addition facts?
    A. Copying of facts.
    B. Explaining the facts
    C. Memorizing the facts.
    D. Reciting the facts.

25. When designing an assessment tool, which of the following is most important to ensure validity?
    A. The tool must measure the same level at any given time.
    B. The tool should measure exactly what it is intended to measure
    C. The assessment should be very difficult and interesting at the same time
    D. The assessment should help the teacher report appropriately.

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SECTION B – Answer any three questions

1a. State four (4) and explain any two (2) differences between the Standard-based curriculum and the Objective-based curriculum. **[16 marks]**

1b. Describe the domains of the National Teachers Standards **[9 marks]**

2a. State four reasons why we teach and learn mathematics **[8 marks]**

2b. Explain the two main types of assessment **[8 marks]**

2c. Justify three important concepts in mathematics which you would expect a child to have understood before introduced to operations on whole numbers. **[9 marks]**

3a. Find the value of the variables P, R and T in the table below. **[6 marks]**

	4	0	R
	P	1	3
+	7	2	5
	1 / 9	/ 3	1 / 4
1	9	T	4

3b. Kofi and Ama worked 43-28 and 378-26 respectively as:

$$\begin{array}{r} 43 \\ -\,\underline{28} \\ \underline{25} \end{array} \quad \text{and} \quad \begin{array}{r} 378 \\ -\,\underline{26} \\ \underline{118} \end{array}$$

[19 marks]

i. Explain their thinking process

ii. Name two appropriate materials you can use to solve the problem

iii. Use the named material to solve the problem through a step-by-step approach.

4a. What is assessment in mathematics **[4 marks]**

4b. Explain assessment as learning, for learning and of learning in mathematics education **[12 marks]**

4c. Describe briefly how you will explain the difference between the terms 'Cardinal number' and 'Ordinal number' to primary class 2 learners. Give an example each to support your explanation. **[9 marks]**

5a. Describe the properties of solids by filling the table **[18 Marks]**

Name of Solid Shape	Number of Faces	Number of Vertices	Number of Edges
Cube			
Cuboids			
Triangular prism			
Square pyramid			
Rectangular Pyramid			
Triangular pyramid			

5b. You are preparing for your micro lesson as an early grade teacher. Explain three relevant things you would do before the lesson and why? **[7Marks]**